

Rosen — §5.1: Mathematical Induction

Selected Exercises

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Exercise 25

Prove that if $h > -1$, then $1 + nh \leq (1 + h)^n$ for all nonnegative integers n . This is called *Bernoulli's inequality*.

Exercise 27

Prove that for every positive integer n ,

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1).$$

Exercise 29

In this and the next exercise, H_n denotes the n th harmonic number, $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n}$. Prove that $H_{2^n} \leq 1 + n$ whenever n is a nonnegative integer.

Exercise 30

Prove that

$$H_1 + H_2 + \cdots + H_n = (n+1)H_n - n.$$

Exercise 35

Prove that $n^2 - 1$ is divisible by 8 whenever n is an odd positive integer.

Exercise 37

Prove that if n is a positive integer, then 133 divides $11^{n+1} + 12^{2n-1}$.

Exercise 40

Prove that if $A_1, A_2, \dots, A_n$ and B are sets, then

$$(A_1 \cap A_2 \cap \cdots \cap A_n) \cup B = (A_1 \cup B) \cap (A_2 \cup B) \cap \cdots \cap (A_n \cup B).$$

Exercise 46

Prove that a set with n elements has $n(n-1)(n-2)/6$ subsets containing exactly three elements whenever n is an integer greater than or equal to 3.

Exercise 52

Suppose that m and n are positive integers with $m > n$ and f is a function from $\{1, 2, \dots, m\}$ to $\{1, 2, \dots, n\}$. Use mathematical induction on the variable n to show that f is not one-to-one.

Exercise 53

Use mathematical induction to show that n people can divide a cake (where each person gets one or more separate pieces of the cake) so that the cake is divided fairly, that is, in the sense that each person thinks he or she got at least $(1/n)$ th of the cake. [*Hint*: For the inductive step, take a fair division of the cake among the first k people, have each person divide their share into what this person thinks are $k + 1$ equal portions, and then have the $(k + 1)$ st person select a portion from each of the k people. When showing this produces a fair division for $k + 1$ people, suppose that person $k + 1$ thinks that person i got p_i of the cake where $\sum_{i=1}^k p_i = 1$.]

Exercise 55

A knight on a chessboard can move one space horizontally (in either direction) and two spaces vertically (in either direction) or two spaces horizontally (in either direction) and one space vertically (in either direction). Suppose that we have an infinite chessboard, made up of all squares (m, n) where m and n are nonnegative integers that denote the row number and the column number of the square, respectively. Use mathematical induction to show that a knight starting at $(0, 0)$ can visit every square using a finite sequence of moves. [*Hint*: Use induction on the variable $s = m + n$.]

Exercise 61

Show that

$$[(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \cdots \wedge (p_{n-1} \rightarrow p_n)] \rightarrow [(p_1 \wedge p_2 \wedge \cdots \wedge p_{n-1}) \rightarrow p_n]$$

is a tautology whenever $p_1, p_2, \dots, p_n$ are propositions, where $n \geq 2$.

Exercise 62

Show that n lines separate the plane into $(n^2 + n + 2)/2$ regions if no two of these lines are parallel and no three pass through a common point.

Exercise 63

Let $a_1, a_2, \dots, a_n$ be positive real numbers. The *arithmetic mean* of these numbers is defined by $A = (a_1 + a_2 + \cdots + a_n)/n$, and the *geometric mean* of these numbers is defined by $G = (a_1 a_2 \cdots a_n)^{1/n}$. Use mathematical induction to prove that $A \geq G$.

Exercise 65

Show that if n is a positive integer, then

$$\sum_{\{a_1, \dots, a_k\} \subseteq \{1, 2, \dots, n\}} \frac{1}{a_1 a_2 \cdots a_k} = n.$$

(Here the sum is over all nonempty subsets of the set of the n smallest positive integers.)

Exercise 68

A guest at a party is a *celebrity* if this person is known by every other guest, but knows none of them. There is at most one celebrity at a party, for if there were two, they would know each other. A particular party may have no celebrity. Your assignment is to find the celebrity, if one exists, at a party, by asking only one type of question—asking a guest whether they know a second guest. Everyone must answer your questions truthfully. That is, if Alice and Bob are two people at the party, you can ask Alice whether she knows Bob; she must answer correctly.

Use mathematical induction to show that if there are n people at the party, then you can find the celebrity, if there is one, with $3(n - 1)$ questions. [*Hint*: First ask a question to eliminate one

person as a celebrity. Then use the inductive hypothesis to identify a potential celebrity. Finally, ask two more questions to determine whether that person is actually a celebrity.]

Exercise 71

Suppose there are n people in a group, each aware of a scandal no one else in the group knows about. These people communicate by telephone; when two people in the group talk, they share information about all scandals each knows about. The gossip problem asks for $G(n)$, the minimum number of telephone calls that are needed for all n people to learn about all the scandals.

Prove that $G(n) = 2n - 4$ for $n \geq 4$.

Exercise 73

Show that if $I_1, I_2, \dots, I_n$ is a collection of open intervals on the real number line, $n \geq 2$, and every pair of these intervals has a nonempty intersection, that is, $I_i \cap I_j \neq \emptyset$ whenever $1 \leq i \leq n$ and $1 \leq j \leq n$, then the intersection of all these sets is nonempty, that is, $I_1 \cap I_2 \cap \dots \cap I_n \neq \emptyset$.